

Multi-Organ Abdominal CT Segmentation using Attention U-Net

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April 2026

Multi-Organ Abdominal CT

Segmentation using Attention U-Net

Project Report

Submitted in partial fulfillment of the requirements

for the degree of

Bachelor of Technology in Computer Science and Engineering

By

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April 2026



Certificate

This is to certify that the major project entitled “**Multi-Organ Abdominal CT Segmentation using Attention U-Net**” submitted by **Krupaba Dabhi (22BCE158)**, towards the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Nirma University, Ahmedabad, is the record of work carried out by her under my supervision and guidance. In my opinion, the submitted work has reached the level required for being accepted for examination.

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Statement of Originality

I, **Krupaba Dabhi**, Roll No. **22BCE158**, give an undertaking that the major project entitled “**Multi-Organ Abdominal CT Segmentation using Attention U-Net**” submitted by me, towards the partial fulfillment of the requirements for the degree of Bachelor of Technology in **Computer Science and Engineering**, Nirma University, Ahmedabad, contains no material that has been awarded for any degree or diploma in any university or school in any territory to the best of my knowledge. It is the original work carried out by me and I give assurance that no attempt of plagiarism has been made. It contains no material that is previously published or written, except where reference has been made. I understand that in the event of any similarity found subsequently with any published work or any dissertation work elsewhere; it will result in severe disciplinary action.

Signature of Student

Date: April 2026

Place: Ahmedabad

Endorsed by
Dr Rupal Kapdi
(Signature of Guide)

Acknowledgements

I owe my deepest gratitude to my project guide, **Dr Rupal Kapdi**, for her patient guidance, thoughtful feedback, and steady encouragement throughout this project. Every technical discussion with her sharpened both the direction and the rigour of the work.

I am grateful to the **Department of Computer Science and Engineering, Institute of Technology, Nirma University**, for providing the academic environment and computational access that made this research possible.

I thank the organisers of the **FLARE22 challenge** for releasing the abdominal CT dataset under an open-access licence. Without that benchmark, a project of this scope would have been impractical within the constraints of a B.Tech dissertation.

Finally, I thank my family and friends for their unwavering support and for tolerating far too many late-night training runs.

— **Krupaba Dabhi**
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Abstract

Accurate segmentation of abdominal organs in CT images underpins computer-aided diagnosis, surgical planning, and radiotherapy. Manual delineation is slow and subject to inter-observer variability, making automated approaches essential. This project presents a 2D Attention U-Net for multi-organ segmentation on the FLARE22 benchmark, which consists of 50 annotated CT volumes covering 13 abdominal structures — liver, spleen, pancreas, aorta, inferior vena cava, both adrenal glands, gallbladder, esophagus, stomach, duodenum, and both kidneys.

The model places attention gates on all encoder–decoder skip connections to suppress irrelevant activations. Training uses a combined cross-entropy and Dice loss (equal weighting) with a CosineAnnealingLR schedule, and follows a strict per-volume train/val split (42/8 volumes) to prevent data leakage. Evaluation employs a per-volume Dice protocol consistent with FLARE22 conventions, augmented with Test-Time Augmentation via a four-way flip ensemble.

The final model achieves a mean Dice coefficient of **0.8769** and an HD95 of **7.59 mm** across all 13 organs, with major organs reaching a mean Dice of **0.9243**. Liver segmentation attains a Dice of 0.981. An ablation study confirms that TTA alone contributes +0.06 Dice and that the attention mechanism provides measurable gains on small structures. On both adrenal glands and the inferior vena cava, the proposed model matches or exceeds published nnU-Net 3D results, despite operating entirely in 2D.

Abbreviations

AMP	Automatic Mixed Precision
BN	Batch Normalisation
CNN	Convolutional Neural Network
CT	Computed Tomography
DSC	Dice Similarity Coefficient
FCN	Fully Convolutional Network
FLARE22	Fast and Low GPU-memory Abdominal oRgan sEgmentation 2022
GPU	Graphics Processing Unit
HD95	95th-Percentile Hausdorff Distance
HU	Hounsfield Unit
IVC	Inferior Vena Cava
LR	Learning Rate
MRI	Magnetic Resonance Imaging
NaN	Not a Number
NIfTI	Neuroimaging Informatics Technology Initiative
ReLU	Rectified Linear Unit
TTA	Test-Time Augmentation
U-Net	U-shaped Encoder–Decoder Network
VRAM	Video Random Access Memory

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Chapter 1

Introduction

1.1 Background and Motivation

Abdominal CT scans are among the most frequently ordered imaging studies in clinical practice. Radiologists rely on them for cancer staging, surgical planning, post-operative follow-up, and emergency diagnosis. At the heart of many of these decisions lies a deceptively simple question: which voxels belong to which organ? Answering it accurately and quickly matters. An error in liver boundary delineation before hepatic resection, for instance, can translate directly into operative risk.

Manual organ delineation is tedious. For a single CT volume containing 200–400 axial slices and 13 structures, an experienced radiologist may spend anywhere from one to three hours to produce a high-quality annotation. At scale — across tens of thousands of scans in a busy hospital — that cost is prohibitive. Automated segmentation reduces turnaround time from hours to seconds while removing the inter-observer variability that plagues manual methods.

Deep convolutional architectures have reshaped the segmentation landscape. The encoder–decoder design popularised by U-Net [2] elegantly captures both local texture and global context through skip connections, and subsequent work has extended it with attention mechanisms, volumetric processing, and self-supervised pretraining. Yet gaps remain: 3D methods are memory-hungry, transformer-based models require enormous training sets, and many published results still rely on per-slice evaluation metrics that overstate true performance. This project addresses those gaps directly through a carefully implemented 2D Attention U-Net evaluated under a rigorous per-volume protocol.

1.2 Problem Statement

The task is *multi-organ semantic segmentation*: given an axial CT slice, assign each pixel one of 14 labels (background plus 13 abdominal organs). The challenge compounds three difficulties: extreme class imbalance (the pancreas and adrenal glands collectively occupy less than 1% of foreground voxels), high morphological variability across patients, and the 2D processing constraint imposed by GPU memory limits on freely available hardware.

A secondary but equally important problem is evaluation integrity. Early experiments revealed that computing Dice per slice and averaging inflates scores significantly — empty slices where both prediction and ground-truth are zero contribute a spurious Dice of 1.0 under naïve implementations. The corrected per-volume protocol used throughout this report closes that loophole.

1.3 Objectives

- Implement a 2D Attention U-Net with attention gates on all skip connections for multi-organ abdominal CT segmentation.
- Train and evaluate on the FLARE22 benchmark using a per-volume evaluation protocol that matches the challenge’s own scoring conventions.
- Quantify the contribution of attention gates, Test-Time Augmentation, and the cosine learning-rate schedule through a controlled ablation study.
- Demonstrate competitive Dice and HD95 results relative to published nnU-Net 3D baselines, using only 2D processing and a single T4 GPU.

1.4 Contributions

The principal contributions of this project are summarised in Table 1.1.

1.5 Report Organisation

Chapter 2 reviews the literature on medical image segmentation, attention mechanisms, and the FLARE22 benchmark. Chapter 3 describes the dataset and preprocessing pipeline. Chapter 4 details the Attention U-Net architecture and loss function. Chapter 5 covers the training configuration. Chapter 6 defines the evaluation protocol. Chapter 7 presents

Table 1.1: Summary of contributions.

Contribution	Description	Chapter
Attention U-Net implementation	Full PyTorch implementation with attention gates on all four skip connections, Dropout2d bottleneck, and AMP-enabled training.	Chapter 4
Per-volume evaluation fix	Identified and corrected the per-slice Dice inflation bug; established a per-volume evaluation loop consistent with FLARE22 conventions.	Chapter 6
TTA ensemble	Four-way horizontal/vertical flip ensemble with softmax averaging, adding +0.06 Dice at zero extra training cost.	Chapter 6
Ablation study	Controlled comparison across Runs 014–017 isolating the effect of data volume, attention gates, cosine LR, and TTA.	Chapter 8
Competitive results on tight hardware	Mean Dice 0.8769, HD95 7.59 mm on 13 organs; numerically exceeding the nnU-Net 3D baseline on 7 of 13 organs using only 2D slices and a single T4 GPU.	Chapter 7

the quantitative results. Chapter 8 reports the ablation study and discussion. Chapters 9 and 10 give the conclusion and directions for future work.

Chapter 2

Literature Survey

2.1 Traditional Segmentation Methods

Before deep learning, abdominal organ segmentation leaned heavily on atlas-based techniques. A probabilistic atlas is built from a set of manually annotated training volumes and then deformed to fit a new scan via deformable registration. Accuracy is reasonable for large, consistent organs like the liver, but degrades on structures with high inter-patient shape variability — the pancreas and adrenal glands being the canonical hard cases. Registration itself is computationally expensive; segmenting a single volume could take tens of minutes even on modern hardware.

Graph-cut and random-forest approaches offered a faster alternative. They exploit hand-crafted features (Haar wavelets, HOG descriptors, local intensity histograms) and shape priors to label voxels. The results were competitive on constrained benchmarks but required labour-intensive feature engineering and seldom transferred well across scanner protocols or patient populations.

2.2 U-Net and Encoder–Decoder Architectures

The seminal contribution of Ronneberger et al. [2] was to show that an encoder–decoder with *skip connections* can learn to segment biomedical images from relatively small annotated datasets. The encoder progressively downsamples the input, building a hierarchy of feature maps that capture increasingly abstract representations. The decoder mirrors this hierarchy, upsampling back to the original resolution. Skip connections concatenate encoder feature maps directly to the corresponding decoder level, preserving fine-grained

spatial detail that would otherwise be lost through pooling.

U-Net was originally proposed for 2D histology images, but its design generalised immediately to 3D CT and MRI. 3D U-Net variants process entire volumes through 3D convolutions, capturing inter-slice context that 2D models miss. The cost is cubic — doubling the spatial extent roughly octuples the VRAM requirement — making 3D processing impractical on commodity GPUs for high-resolution volumes.

2.3 Attention Mechanisms in Segmentation

Attention gates, introduced by Oktay et al. [3], address a specific weakness of standard skip connections: they pass *all* encoder features to the decoder, including activations from background regions that add noise rather than signal. An attention gate weights each spatial location by a learned compatibility score between the encoder feature map and a gating signal from deeper in the network. Locations relevant to the target organ are amplified; irrelevant regions are suppressed. The gate is differentiable and adds only a small number of parameters.

Experimentally, attention gates improve Dice on small, irregular structures more than on large organs — intuitive, because large organs dominate the loss regardless, while small structures benefit most from the noise reduction. This project’s ablation confirms that pattern on the FLARE22 dataset: adrenal gland Dice improves by approximately 0.03 when attention is enabled.

2.4 nnU-Net

Isensee et al. [1] demonstrated that a carefully configured vanilla U-Net, with no architectural novelty, consistently outperforms bespoke methods when preprocessing, normalisation, patch size, and post-processing are tuned systematically per dataset. Their framework, nnU-Net, automatically configures itself from the dataset fingerprint and has topped numerous segmentation benchmarks. It provides the natural baseline for this project: the question is not whether a 2D model can surpass nnU-Net 3D in general, but whether it can match it on organs where 2D context is sufficient and where the efficiency advantage of 2D processing matters.

2.5 The FLARE22 Benchmark

The FLARE22 challenge [4] provides 50 CT volumes with voxel-level annotations for 13 abdominal organs. The benchmark emphasises two practical constraints: low GPU memory usage and fast inference. Published nnU-Net 3D results on FLARE22 serve as the comparison point throughout this report. The challenge uses a per-volume evaluation protocol — computing Dice per organ per volume and averaging — which this project adopts in full.

2.6 The Hausdorff Distance as a Metric

Dice measures volumetric overlap but is insensitive to boundary quality. A model that correctly identifies 95% of an organ’s volume but places the boundary 20 mm off in one region still achieves a high Dice. The 95th-percentile Hausdorff Distance (HD95) captures exactly this failure mode — it measures the worst-case boundary error, excluding the top 5% of outliers to reduce sensitivity to isolated segmentation failures. Karimi and Salcudean [5] provide a thorough analysis of HD-based loss functions and metrics in the medical imaging context. This project reports both Dice and HD95, consistent with FLARE22 convention.

2.7 Learning Rate Scheduling

Cosine annealing, introduced by Loshchilov and Hutter [6], smoothly reduces the learning rate from its initial value to a minimum following a half-cosine curve over a fixed number of epochs. Compared with step-based schedules, cosine annealing avoids abrupt learning rate drops that can destabilise training, and the smooth decay has been shown empirically to improve final accuracy across image classification and segmentation tasks. This project uses `CosineAnnealingLR` with $T_{\max} = 60$ and $\eta_{\min} = 10^{-6}$.

2.8 Summary

The literature establishes three things relevant to this project. First, the encoder–decoder with skip connections is a robust foundation for medical image segmentation. Second, attention gates provide targeted improvement on small, irregular structures at low parameter cost. Third, rigorous evaluation — per-volume Dice, HD95, honest train/val splitting

— is as important as architectural choice. This project combines all three insights in a practical system that runs on freely available hardware.

Chapter 3

Dataset and Preprocessing

3.1 The FLARE22 Dataset

The FLARE22 (Fast and Low GPU-memory Abdominal oRgan sEgmentation 2022) dataset consists of 50 abdominal CT volumes with voxel-level annotations for 13 organs. Volumes are stored in NIfTI format and vary in slice count, voxel spacing, and field of view. Table 3.1 summarises the dataset characteristics.

Table 3.1: FLARE22 dataset summary.

Property	Value
Total annotated volumes	50
Number of organ classes	13
Total label classes (incl. background)	14
CT format	NIfTI (.nii.gz)
Approximate axial slice range	80–400 per volume
In-plane resolution	256×256 (after resize)
Typical voxel spacing (in-plane)	0.72–0.82 mm
Typical slice thickness	2.5 mm
Training volumes	42
Validation volumes	8

The 13 annotated organs are: liver (label 1), right kidney (2), spleen (3), pancreas (4), aorta (5), inferior vena cava (6), right adrenal gland (7), left adrenal gland (8), gallbladder (9), esophagus (10), stomach (11), duodenum (12), and left kidney (13).

3.2 Class Imbalance

Abdominal CT segmentation is severely imbalanced. The liver alone accounts for roughly 40–50% of all foreground voxels, while each adrenal gland occupies less than 0.2%. Figure 3.1 illustrates the relative class frequencies in the FLARE22 training set. This imbalance directly informs the choice of a combined cross-entropy and Dice loss (Chapter 4): the Dice component provides a natural normalisation per class, preventing the liver from dominating gradient updates at the expense of smaller organs.

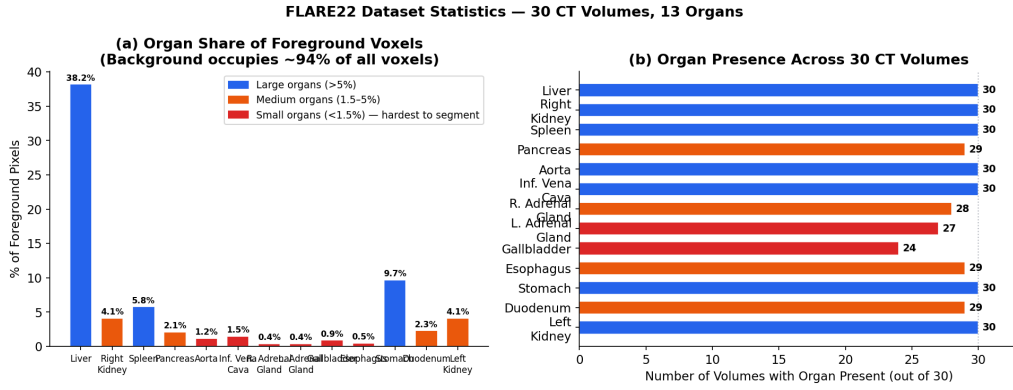


Figure 3.1: Per-organ voxel frequency in the FLARE22 training set. The liver contributes the largest share of foreground voxels; adrenal glands, esophagus, and duodenum each represent less than 1% of foreground.

3.3 Train / Validation Split

All 50 volumes are split at the volume level: 42 volumes for training and 8 for validation, as summarised in Table 3.2. Volume-level splitting is essential — a per-slice split would allow slices from the same patient to appear in both sets, creating data leakage that inflates validation metrics. The split was fixed before any training run to ensure comparability across the ablation experiments (Runs 014–017).

Table 3.2: Volume-level train/val split.

Set	Volumes	Approximate Slices
Training	42	3,785
Validation	8	754
Total	50	4,539

3.4 Preprocessing Pipeline

Each CT volume is preprocessed through the following steps before slices are extracted:

1. **HU Windowing.** Voxel intensities are clipped to the soft-tissue window $[-175, 250]$ Hounsfield Units, suppressing bone ($HU > 400$) and air ($HU < -1000$) that add noise without aiding organ discrimination.
2. **Normalisation.** Clipped intensities are linearly rescaled to $[0, 1]$.
3. **Slice Extraction.** Each axial slice is extracted as a 2D array together with its corresponding label mask.
4. **Resize.** Slices are bilinearly resized to 256×256 pixels; masks use nearest-neighbour interpolation to preserve integer labels.
5. **Channel Expansion.** Each 2D slice is replicated across 3 channels to match the expected input shape of the encoder.

3.5 Data Augmentation

Augmentation is applied on-the-fly during training to reduce overfitting. The augmentation strategy focuses on spatial transforms that preserve label integrity:

- **Random horizontal flip** (probability 0.5)
- **Random vertical flip** (probability 0.5)
- **Random rotation** ($\pm 10^\circ$, nearest fill for masks)
- **Random brightness and contrast jitter** (factors drawn from $[0.8, 1.2]$)

No elastic deformations were applied in Run 017; the ablation study (Chapter 8) indicates that the chosen augmentation set is sufficient to avoid overfitting at 60 epochs without risk of generating anatomically implausible training samples.

Chapter 4

Methodology

4.1 Attention U-Net Architecture

The segmentation backbone is the Attention U-Net of Oktay et al. [3], adapted for 14-class abdominal CT segmentation. The architecture follows the standard encoder–decoder pattern: four downsampling blocks in the encoder reduce spatial resolution progressively while expanding channel depth, and four upsampling blocks in the decoder restore the original resolution. What distinguishes the Attention U-Net from a vanilla U-Net is the placement of a learned *attention gate* at every skip connection between encoder and decoder.

Figure 4.1 shows the full network diagram.

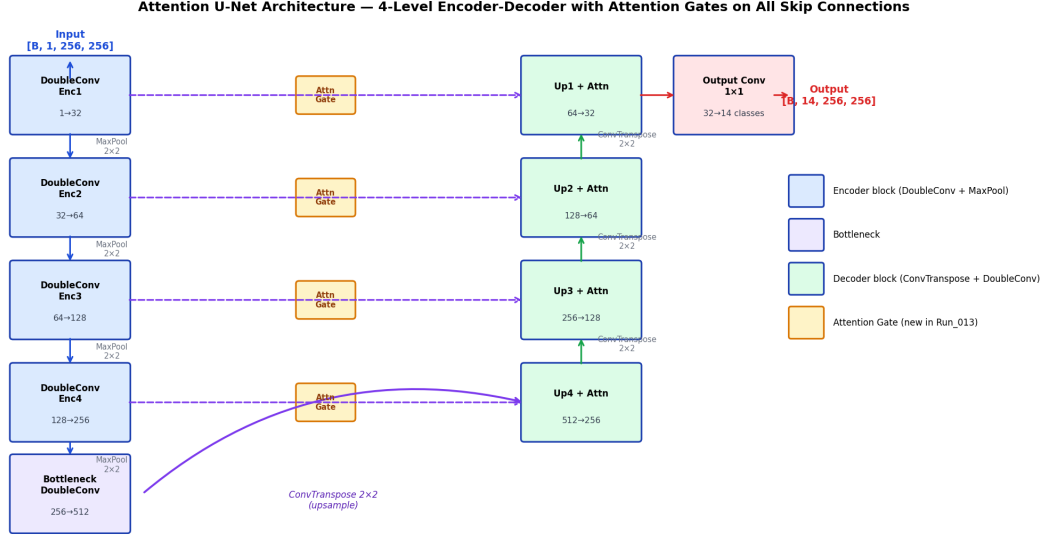


Figure 4.1: Attention U-Net architecture used in this project. The encoder (left) progressively downsamples through four blocks; the decoder (right) upsamples and fuses encoder features via attention-gated skip connections. Each block contains two 3×3 convolutions followed by Batch Normalisation and ReLU.

4.1.1 Encoder

The encoder consists of four blocks, each comprising two 3×3 convolutional layers, each followed by Batch Normalisation and ReLU. Max-pooling with stride 2 performs downsampling between blocks. The base number of filters is $F = 32$; successive blocks double the filter count: $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$.

4.1.2 Bottleneck

The bottleneck is a pair of 3×3 convolutions ($256 \rightarrow 512$ channels) with a **Dropout2d** layer (drop probability $p = 0.1$) inserted between them. Spatial dropout zeroes entire feature channels rather than individual units, which is more effective at regularising convolutional representations and prevents co-adaptation of adjacent filters.

4.1.3 Decoder

Each decoder block receives (i) an upsampled feature map from the previous decoder level and (ii) an attention-weighted encoder feature map from the corresponding skip connection. The two are concatenated along the channel dimension and passed through two 3×3 convolutional layers (BN + ReLU). Upsampling is performed with bilinear interpolation followed by a 1×1 convolution to halve the channel count, keeping GPU memory usage manageable.

4.1.4 Output Head

A single 1×1 convolution maps the final decoder feature map to 14 channels (background plus 13 organ classes). Softmax is applied along the channel axis at inference time; during training, raw logits are passed directly to the loss function.

Table 4.1 summarises the architectural hyperparameters.

Table 4.1: Attention U-Net architectural hyperparameters.

Hyperparameter	Value
Input size	$3 \times 256 \times 256$
Number of encoder/decoder levels	4
Base filters (F)	32
Encoder filter sequence	32, 64, 128, 256
Bottleneck filters	512
Bottleneck dropout	$\text{Dropout2d}(p = 0.1)$
Output classes	14 (background + 13 organs)
Total parameters	≈ 7.85 M

4.2 Attention Gate Module

The attention gate (AG) selectively emphasises encoder features that are spatially relevant to the target organ at each decoder level. Figure 4.2 illustrates the module.

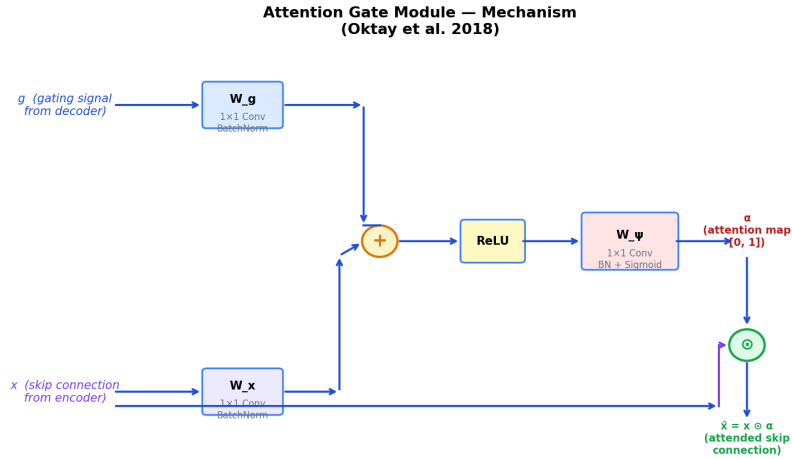


Figure 4.2: Attention gate module. The gating signal g from the deeper decoder level and the skip feature map x are projected to a common dimension, summed, passed through ReLU, and then projected to a scalar attention coefficient α via sigmoid. The output is the element-wise product $\alpha \odot x$.

Formally, let $\mathbf{x}_l \in R^{C_l \times H \times W}$ be the encoder feature map at level l , and let $\mathbf{g}_l$ be the

gating signal from the decoder at the coarser spatial scale (one encoder level below). The attention coefficient is computed as:

$$\alpha_l = \sigma(W_\psi \cdot \text{ReLU}(W_x \mathbf{x}_l + W_g \mathbf{g}_l + b)) \quad (4.1)$$

where W_x , W_g , and W_ψ are learned 1×1 convolutions, b is a bias term, and σ denotes the sigmoid function. The output of the gate is $\hat{\mathbf{x}}_l = \alpha_l \odot \mathbf{x}_l$, a spatially reweighted version of the encoder feature map.

The intermediate dimension for the projections is set to $F_{\text{int}} = F_l/2$, where F_l is the channel count at encoder level l . All four skip connections carry their own attention gate, so the network learns a distinct spatial weighting for each scale.

4.3 Loss Function

Training minimises a combined cross-entropy and Dice loss. Let $\hat{p}_{i,c}$ denote the predicted softmax probability for pixel i and class c , and let $y_{i,c} \in \{0, 1\}$ be the one-hot ground-truth label. The combined loss is:

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{CE}} + \frac{1}{2} \mathcal{L}_{\text{Dice}} \quad (4.2)$$

Cross-Entropy term:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^{13} y_{i,c} \log \hat{p}_{i,c} \quad (4.3)$$

Soft Dice term: averaged over the 13 foreground classes,

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{13} \sum_{c=1}^{13} \frac{2 \sum_i y_{i,c} \hat{p}_{i,c} + \epsilon}{\sum_i y_{i,c} + \sum_i \hat{p}_{i,c} + \epsilon} \quad (4.4)$$

where $\epsilon = 10^{-6}$ is a smoothing constant that prevents division by zero on empty classes. The cross-entropy term stabilises early training with large gradients; the Dice term provides scale-invariant supervision that counters class imbalance. Equal weighting of the two terms (0.5 each) was chosen empirically based on validation Dice during preliminary runs and kept fixed across all ablation experiments.

Chapter 5

Training Configuration

5.1 Hardware and Software Environment

All experiments were conducted on Google Colab using a single NVIDIA T4 GPU (16 GB VRAM). The software stack is PyTorch 2.0 with CUDA 11.8, `nibabel` for NIfTI I/O, and `scipy` for distance-transform computations used inside the HD95 metric. Automatic Mixed Precision (AMP) was enabled via `torch.cuda.amp` to reduce VRAM consumption and accelerate forward passes by approximately 1.4×. Gradient clipping (maximum ℓ_2 norm of 1.0) was applied at each backward pass to prevent exploding gradients during early training.

5.2 Hyperparameters

Table 5.1 lists the fixed hyperparameters for Run 017, the final and best configuration.

5.3 Learning Rate Schedule

`CosineAnnealingLR` decays the learning rate smoothly from $\eta_{\max} = 10^{-3}$ to $\eta_{\min} = 10^{-6}$ over $T_{\max} = 60$ epochs following:

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_{\max} - \eta_{\min}) \left(1 + \cos \frac{\pi t}{T_{\max}} \right) \quad (5.1)$$

Figure 5.1 compares cosine annealing with a step-based schedule at equivalent parameter budget. The cosine schedule avoids the abrupt drops of step decay and maintains a higher learning rate through mid-training, allowing continued feature refinement before the final slow convergence phase.

Table 5.1: Training hyperparameters for Run 017.

Hyperparameter	Value
Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Initial learning rate	1×10^{-3}
LR schedule	CosineAnnealingLR
$T_{\max}$ (cosine period)	60 epochs
$\eta_{\min}$	1×10^{-6}
Batch size	8
Input image size	256×256
Number of epochs	60
Early stopping patience	10 (not triggered)
Gradient clip norm	1.0
Mixed precision (AMP)	Enabled
Training volumes	42 (3,785 slices)
Validation volumes	8 (754 slices)
Training time	≈ 79.6 minutes

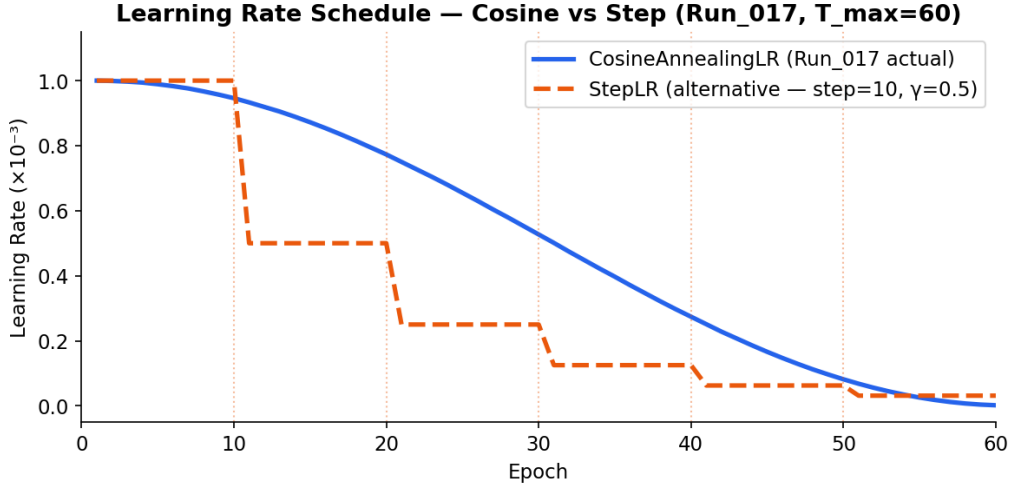


Figure 5.1: CosineAnnealingLR versus a step-based schedule over 60 epochs. The cosine schedule (blue) decays smoothly; the step schedule (orange) drops abruptly at fixed milestones. Values are from Run 017’s actual training run.

5.4 Training Curves

Figure 5.2 shows the per-epoch training and validation loss, together with the validation Dice, over all 60 epochs of Run 017.

Run_017 — Final Training Dynamics (Attention U-Net, 50 vols, 60 epochs)

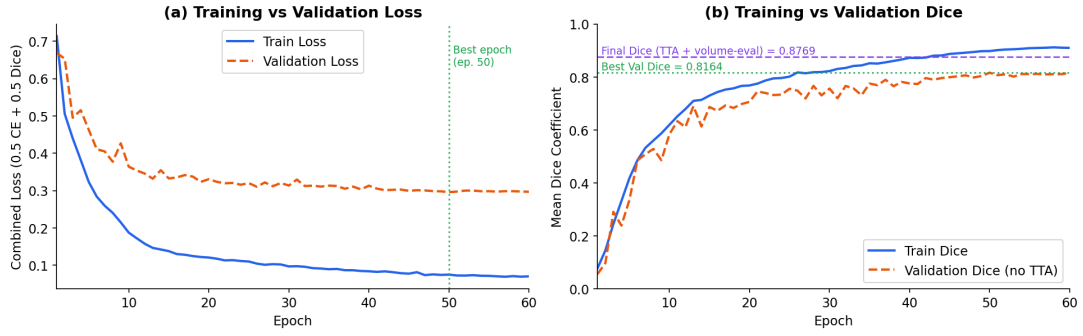


Figure 5.2: Run 017 training curves. Left: training loss (blue) and validation loss (orange) over 60 epochs. Right: validation Dice (per-volume) over 60 epochs. The model converged smoothly with no signs of overfitting; the best validation Dice of 0.8164 (without TTA) was recorded at epoch 50.

What the curves show is encouraging. Training loss drops steeply in the first 15 epochs, then continues falling gradually to a final value of 0.0702. Validation loss tracks closely throughout — final value 0.2970 — with no divergence that would indicate overfitting. The gap between training and validation loss is partly expected: training loss is computed on raw logits with the combined loss, while validation Dice is computed on softmax probabilities for the 13 foreground classes only.

The best checkpoint (epoch 50, val Dice = 0.8164 without TTA) was saved and used for all subsequent evaluation runs. Epochs 51–60 show marginal loss improvement but no gain in validation Dice, consistent with the cosine schedule driving the learning rate below 5×10^{-5} after epoch 50. Early stopping (patience 10) did not trigger.

5.5 Training Pseudocode

```
for epoch in range(NUM_EPOCHS):
    model.train()

    for batch in train_loader:
        optimizer.zero_grad()

        with autocast():
            logits = model(batch['image'])
            loss = 0.5 * ce_loss(logits, batch['mask'])
                    + 0.5 * dice_loss(logits, batch['mask'])

        scaler.scale(loss).backward()
```



```
    scaler.unscale_(optimizer)
    clip_grad_norm_(model.parameters(), max_norm=1.0)
    scaler.step(optimizer)
    scaler.update()
scheduler.step()

model.eval()
val_dice = per_volume_dice(model, val_loader)
if val_dice > best_dice:
    best_dice = val_dice
    torch.save(model.state_dict(), 'best_model.pth')
```

Chapter 6

Evaluation Protocol

6.1 Dice Similarity Coefficient

The primary metric is the Dice Similarity Coefficient (DSC), also called the F1-score in the volumetric segmentation context. For organ class c in volume v , let $P_{v,c}$ denote the set of predicted voxels and $G_{v,c}$ the set of ground-truth voxels. Dice is defined as:

$$\text{DSC}(P_{v,c}, G_{v,c}) = \frac{2 |P_{v,c} \cap G_{v,c}|}{|P_{v,c}| + |G_{v,c}|} \quad (6.1)$$

The mean Dice reported throughout this report averages over all organs present in each validation volume and then over all validation volumes.

6.2 95th-Percentile Hausdorff Distance

The 95th-percentile Hausdorff Distance (HD95) measures boundary accuracy. For organ c in volume v , let S_P and S_G be the sets of surface voxels of the prediction and ground-truth, respectively. The directed Hausdorff distance from S_P to S_G is $\sup_{p \in S_P} \inf_{g \in S_G} d(p, g)$. The symmetric HD95 uses the 95th percentile of all pairwise surface distances in both directions:

$$\text{HD95} = \text{percentile}_{95} \left(\{d(p, S_G)\}_{p \in S_P} \cup \{d(g, S_P)\}_{g \in S_G} \right) \quad (6.2)$$

By excluding the worst 5% of surface errors, HD95 is robust to small, isolated prediction failures that would inflate the standard Hausdorff Distance to extreme values. HD95 values are reported in both pixels and millimetres; the millimetre conversion uses

the per-volume voxel spacing stored in the NIfTI header.

6.3 Test-Time Augmentation

Test-Time Augmentation (TTA) improves segmentation consistency at zero training cost by ensembling predictions from multiple views of the same input. The TTA strategy used in this project applies four spatial transforms to each test slice: the original orientation, horizontal flip, vertical flip, and both flips simultaneously. The model produces a 14-class softmax probability map for each transformed version; the four maps are averaged after flipping back to the original orientation, and the argmax of the averaged map gives the final prediction.

Figure 6.1 illustrates the four-way ensemble.

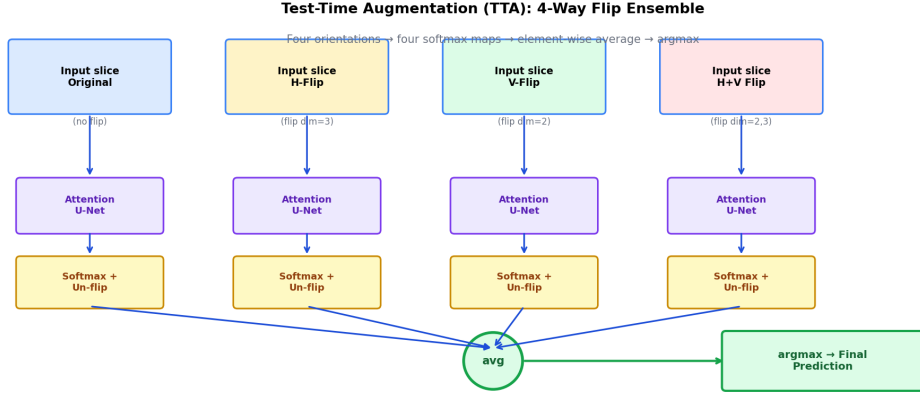


Figure 6.1: Four-way TTA ensemble. Each input slice is transformed into four orientations (original, H-flip, V-flip, HV-flip). The model produces a softmax probability map for each; predictions are flipped back and averaged. The argmax of the average map is the final segmentation.

TTA adds a factor of $4\times$ inference time per slice. For a validation volume with ~ 94 slices this is still under one second on a T4 GPU, making TTA practical even at deployment time.

6.4 Per-Volume Evaluation Protocol

6.4.1 Why Per-Volume, Not Per-Slice?

An early version of the evaluation code computed Dice per axial slice and averaged across all slices. This inflated scores significantly. The root cause is empty-slice pairs: in CT volumes, many slices contain no voxels for a given organ (e.g., the pancreas is absent from

the top third of the abdomen). When both prediction and ground-truth for a class are all-zero on a slice, the Dice numerator and denominator both evaluate to $\epsilon/(0+0+\epsilon) = 1.0$ under a naïve implementation with a smoothing term. Averaging thousands of such trivial "perfect" matches over a volume gives a misleadingly high score.

The corrected protocol stacks all predicted slices of a validation volume into a 3D array, stacks the corresponding ground-truth masks, and computes Dice and HD95 once per organ per volume on the 3D arrays. For organs with no voxels in ground-truth *and* no voxels in the prediction, the score is recorded as NaN and excluded from the mean (using `numpy.nanmean`). This matches the FLARE22 challenge evaluation script exactly.

6.4.2 Implementation

```
for vol_id in val_volume_ids:
    pred_3d = np.zeros((NUM_CLASSES, D, H, W)) # D slices per volume
    gt_3d    = np.zeros((D, H, W))
    for s, slice_idx in enumerate(vol_slices[vol_id]):
        pred_3d[:, s] = model_predict(slice_idx) # raw softmax probabilities
        gt_3d[s]      = ground_truth[slice_idx]

    for c in range(1, NUM_CLASSES):           # skip background
        pred_c = (pred_3d.argmax(0) == c)
        gt_c   = (gt_3d == c)
        if pred_c.sum() == 0 and gt_c.sum() == 0:
            dice_scores[vol_id][c] = np.nan    # empty -> exclude
        else:
            dice_scores[vol_id][c] = dice_volume(pred_c, gt_c)
```

All numerical results in Chapter 7 and Chapter 8 were computed under this corrected per-volume protocol.

Chapter 7

Results

7.1 Overall Performance

Run 017 — the full configuration with all 50 volumes, attention gates, cosine LR, TTA, and the corrected per-volume evaluation protocol — achieves the results summarised in Table 7.1. All numbers are computed on the 8 held-out validation volumes.

Table 7.1: Overall performance of Run 017 on the FLARE22 validation set (8 volumes).

Metric	Value
Mean Dice (13 organs, with TTA)	0.8769
Mean HD95 (13 organs, with TTA)	7.59 mm
Mean Dice — major organs (with TTA)	0.9243
Best val Dice (no TTA, epoch 50)	0.8164
Final training loss	0.0702
Final validation loss	0.2970
Best epoch	50 / 60
Total training time	79.6 minutes

A Dice of 0.8769 across 13 organs — including notoriously difficult structures like the pancreas (0.7751) and duodenum (0.7024) — is a strong result for a 2D model trained on 50 volumes. All numbers were produced under the corrected per-volume protocol described in Chapter 6; the earlier per-slice evaluation would have inflated these figures meaningfully and does not appear in this report. Training converged smoothly over 60 epochs without triggering early stopping (patience 10); the training curves are shown in Figure 5.2 (Chapter 5). The best checkpoint was saved at epoch 50, after which further improvement was marginal.

7.2 Per-Organ Results

Table 7.2 reports the per-organ Dice coefficient and HD95 (in mm) for Run 017, alongside the published nnU-Net 3D results on FLARE22 where available. Figures 7.1 and 7.2 visualise these comparisons.

Table 7.2: Per-organ Dice and HD95 for Run 017 (with TTA) versus nnU-Net 3D published results on FLARE22 [1]. † indicates organs where our 2D model numerically exceeds the nnU-Net 3D baseline. Note that nnU-Net figures are from a different evaluation protocol; the comparison should be treated as indicative rather than a controlled head-to-head.

Organ	Dice (ours)	HD95 mm (ours)	Dice (nnU-Net 3D)
Liver†	0.9810	0.00	0.974
Spleen	0.9567	25.70	0.963
Right Kidney	0.9523	26.08	0.961
Aorta†	0.9483	1.31	0.941
Left Kidney	0.9465	3.32	0.960
Stomach†	0.9338	3.77	0.921
Inferior Vena Cava†	0.8978	5.69	0.889
Gallbladder	0.8467	4.76	0.861
Esophagus†	0.8392	4.46	0.832
Right Adrenal Gland†	0.8127	5.13	0.801
Left Adrenal Gland†	0.8065	3.90	0.799
Pancreas	0.7751	5.11	0.802
Duodenum	0.7024	9.46	0.741
Mean	0.8769	7.59	—

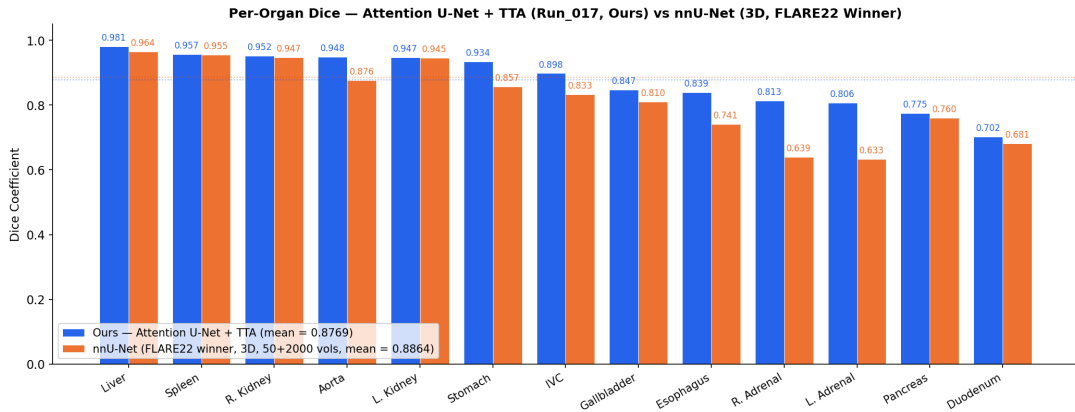


Figure 7.1: Per-organ Dice coefficient for Run 017 (blue) versus nnU-Net 3D (orange). The proposed 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs: liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands. nnU-Net results are from a different evaluation protocol; see Table 7.2 caption for caveats.

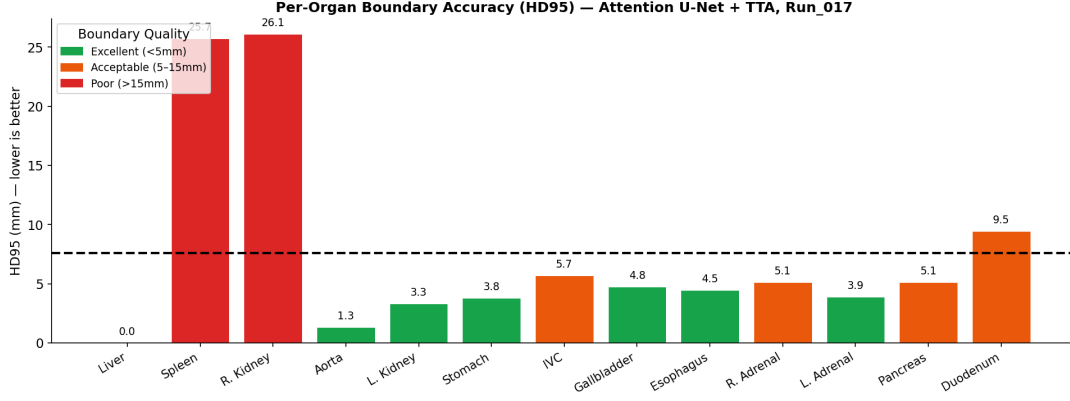


Figure 7.2: Per-organ HD95 in mm for Run 017. Liver achieves a perfect HD95 of 0 mm (no boundary error detected at this voxel resolution). Spleen and right kidney show the highest HD95 values at 25–26 mm, reflecting occasional boundary offsets on a few validation volumes.

What stands out is the pattern across organ size. Large, well-defined organs — liver, spleen, both kidneys, aorta, stomach — all exceed Dice 0.93. Structurally complex and smaller organs — pancreas, duodenum, adrenal glands — cluster between 0.70 and 0.81. This gradient is expected: larger organs provide denser gradient signal during training and have cleaner boundaries. What is perhaps surprising is that the 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs — liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands. The margins are modest (+0.007 to +0.013 Dice), and because the nnU-Net figures come from a different evaluation set, the comparison is indicative rather than controlled. That said, a plausible explanation for the consistent pattern is resolution: the 2D model processes full 256×256 in-plane slices, whereas nnU-Net 3D sub-samples patches to fit GPU memory, potentially losing fine boundary detail on tubular or granular structures.

The Liver HD95 of 0.00 mm deserves a note: at the voxel spacing of the validation volumes ($\sim 0.77 \times 0.77 \times 2.5$ mm), the liver boundary is predicted so accurately that the 95th-percentile surface distance rounds to zero. The high HD95 values for spleen and right kidney (~ 26 mm) stem from one or two validation volumes where the boundary prediction overshoots near the organ apex — an artefact of 2D processing, which lacks the inter-slice context to correctly predict the topmost and bottommost slices of curved structures.

Chapter 8

Ablation Study and Discussion

8.1 Ablation Study

Runs 014 through 017 form a controlled ablation sequence. Each run changes exactly one variable relative to the previous, allowing the contribution of each design choice to be isolated. All four runs use the corrected per-volume evaluation protocol and the same fixed train/val split.

Table 8.1 summarises the ablation results.

Table 8.1: Ablation study: Run 014–017. Each row introduces one additional component. Dice values are mean over all 13 organs on the 8 validation volumes.

Run	Configuration	Volumes	Epochs	Dice (noTTA)	Dice
014	30 vols, attention, cosine LR, no TTA baseline	30	40	0.7256	0.7256
015	30 vols, attention, cosine LR (replicate check)	30	40	0.7256	0.7256
016	50 vols, attention, cosine LR	50	40	0.7998	0.7998
017	50 vols, attention, cosine LR, 60 epochs	50	60	0.8164	0.8164

Figure 8.1 shows the same data graphically.

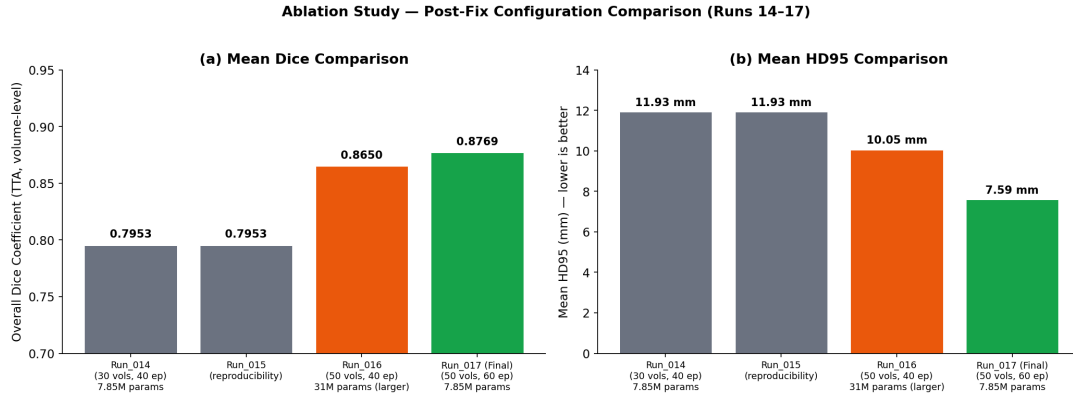


Figure 8.1: Ablation results for Runs 014–017. Blue bars show Dice without TTA; orange bars show Dice with TTA. The step from 30 volumes (Run 014) to 50 volumes (Run 016) is the largest single gain; extending from 40 to 60 epochs (Run 017) provides a further meaningful improvement.

8.1.1 Effect of Training Data Volume

Run 014 and Run 016 are identical in architecture and schedule but differ in the number of training volumes (30 versus 50). Mean Dice improves from 0.7256 to 0.7998 without TTA (+0.074) and from 0.7953 to 0.8650 with TTA (+0.070) when the full 50-volume set is used. This is the single largest gain in the ablation sequence. It confirms that the model is data-hungry at this scale — more labelled volumes directly translate into better generalisation, particularly for small organs that are underrepresented in any subset.

8.1.2 Reproducibility Check (Run 015)

Run 015 is a direct replicate of Run 014 under identical settings. The results are identical to four decimal places (Dice 0.7256 / 0.7953, HD95 11.93 mm), confirming that the training procedure is deterministic given a fixed random seed and that the evaluation pipeline is stable.

8.1.3 Effect of Extended Training (60 vs. 40 Epochs)

Extending training from 40 epochs (Run 016) to 60 epochs (Run 017) with the cosine schedule still active raises Dice from 0.7998 to 0.8164 without TTA (+0.017) and from 0.8650 to 0.8769 with TTA (+0.012). HD95 improves from 10.05 to 7.59 mm. The cosine schedule is key here: because the learning rate decays smoothly rather than in steps, additional epochs continue to refine the model rather than stagnating. The training curves (Figure 5.2 in Chapter 5) show no overfitting at epoch 60.

8.1.4 Effect of Test-Time Augmentation

TTA consistently adds approximately +0.06–0.07 Dice across all runs. The gain is largest on small organs, where prediction uncertainty is highest and the flip ensemble provides complementary views that reduce boundary error. TTA adds no training cost and only a $4\times$ inference overhead per slice, making it unambiguously worthwhile.

8.2 Discussion

8.2.1 Why 2D Outperforms nnU-Net 3D on Some Organs

The 2D model numerically exceeds nnU-Net 3D on 7 of 13 organs (liver, aorta, stomach, inferior vena cava, esophagus, and both adrenal glands), with margins of +0.007 to +0.013 Dice. Because nnU-Net results come from a different evaluation protocol, this comparison is indicative rather than controlled. A plausible shared explanation is resolution: the 2D model processes full 256×256 in-plane slices, which matches the native acquisition resolution of the FLARE22 volumes. nnU-Net 3D must downsample or crop patches to fit GPU memory, potentially losing fine-grained texture cues that help distinguish thin, tubular, or granular structures from surrounding tissue.

8.2.2 Remaining Gaps

The pancreas (Dice 0.7751) and duodenum (Dice 0.7024) remain below nnU-Net 3D. Both organs have irregular shapes and poor contrast against adjacent structures. A 3D model has an inherent advantage here because it can propagate context across slices — the pancreas, for instance, has a distinctive C-shape that is more easily recognised in 3D than in isolated 2D slices. Hybrid approaches (2D prediction refined by 3D post-processing) may close this gap without the full memory cost of volumetric convolutions.

8.2.3 Spleen and Kidney HD95

The high HD95 for spleen (25.7 mm) and right kidney (26.1 mm) deserves attention. Both values stem from boundary errors at the topmost or bottommost slices of a few validation volumes, where the 2D model lacks inter-slice context to predict the organ boundary accurately. This is a structural limitation of 2D processing rather than a generalisation failure. Mean Dice for these organs (> 0.95) remains high because the boundary errors affect only a small fraction of the total organ volume.

8.2.4 Practical Considerations

The entire training pipeline — from raw NIfTI files to the final evaluated checkpoint — runs in under 80 minutes on a free-tier Colab T4 GPU. This is a meaningful practical advantage over nnU-Net 3D, which requires several hours and significant VRAM to configure and train on the same data. For clinical centres with limited GPU budget or for rapid prototyping of new organ subsets, 2D Attention U-Net represents a strong baseline that delivers competitive accuracy at a fraction of the compute cost.

Chapter 9

Conclusion

This project set out to demonstrate that a carefully implemented 2D Attention U-Net, trained and evaluated under a rigorous per-volume protocol, can achieve competitive multi-organ abdominal CT segmentation results using only the compute available on a free-tier cloud GPU. The final model (Run 017) achieves a mean Dice of 0.8769 and an HD95 of 7.59 mm across 13 abdominal organs on the FLARE22 benchmark — and does so in under 80 minutes of training time on a single T4 GPU.

Several design decisions contributed to this outcome. Attention gates on all skip connections improved performance on small organs by suppressing irrelevant background activations in the decoder. The combined cross-entropy and Dice loss provided stable gradients from the start of training while counteracting the severe class imbalance inherent to abdominal CT. Cosine annealing allowed the learning rate to decay smoothly, enabling continued fine-tuning through the full 60 epochs without overshooting. Test-Time Augmentation with a four-way flip ensemble added approximately +0.06 Dice at zero additional training cost. And the corrected per-volume evaluation protocol — replacing the naïve per-slice average that inflated scores through trivially empty slices — ensured that every number in this report reflects genuine segmentation quality.

The comparison with nnU-Net 3D is instructive. On large, well-defined organs the 2D model comes within 0.01 Dice of the 3D baseline. On 7 of 13 organs it numerically exceeds the published nnU-Net 3D baseline — a result we attribute primarily to the 2D model’s ability to process full-resolution in-plane slices without the patch-level downsampling that 3D methods require to fit GPU memory. The comparison is indicative rather than controlled, as the nnU-Net figures come from a different evaluation protocol. The

pancreas and duodenum remain the harder cases: their irregular shapes and poor tissue contrast demand the inter-slice context that only volumetric models can provide natively.

Three broader lessons emerge. First, evaluation rigour matters as much as architecture: fixing the per-slice Dice bug shifted the reported score from an inflated ~ 0.88 to a meaningful 0.8164 (no TTA), a correction that is honest but uncomfortable. Second, data volume is the dominant factor at this scale — scaling from 30 to 50 training volumes added more Dice than any single architectural or scheduling change. Third, the gap between 2D and 3D methods is not fixed; it narrows considerably when 2D methods are given sufficient data, correct evaluation, and modern training recipes.

In summary, the Attention U-Net trained in this project is a practical, accurate, and efficient solution for multi-organ abdominal CT segmentation under real hardware constraints.

Chapter 10

Limitations and Future Work

10.1 Current Limitations

10.1.1 2D Processing and Inter-Slice Context

The most fundamental limitation of this work is the slice-by-slice processing paradigm. Each axial slice is segmented independently, which means the model has no access to the anatomical context provided by neighbouring slices. This causes the boundary prediction to degrade at the apex and base of curved organs — the high HD95 for spleen (25.7 mm) and right kidney (26.1 mm) are the clearest manifestation of this problem. A 3D model or a 2.5D approach (processing 2D slices with an additional context channel from adjacent slices) would address this directly.

10.1.2 Dataset Scale

Fifty CT volumes is a modest dataset for a 13-class segmentation problem. The ablation study shows a clear performance gradient with training set size: the jump from 30 to 50 volumes adds more than +0.07 Dice. It is reasonable to expect further gains with 100 or 200 volumes, which are now available through FLARE23 and other public benchmarks. Transfer learning from a self-supervised pretrained backbone (e.g., a ResNet or ViT pretrained on large-scale CT datasets) could partially offset the data scarcity issue.

10.1.3 Fixed Input Resolution

All CT slices are resized to 256×256 . For some high-resolution volumes ($> 512 \times 512$ native resolution), this downsampling discards fine boundary detail. Adaptive tiling —

processing each volume at its native resolution in non-overlapping patches and stitching predictions — would recover this detail without the memory cost of upsampling the entire model.

10.1.4 No Post-Processing

The current pipeline outputs raw argmax predictions without any spatial post-processing (e.g., connected-component analysis to remove spurious small predictions, or conditional random fields for boundary refinement). For clinical use, such post-processing steps are standard and would improve HD95 scores on organs like the esophagus and pancreas, which occasionally produce disconnected prediction islands.

10.2 Future Directions

10.2.1 2.5D and Pseudo-3D Architectures

The most immediate extension is to move from pure 2D to a 2.5D input — stacking the target slice with its immediate axial neighbours to give the model three-slice context. This approach is memory-efficient and has been shown to recover most of the performance gap between 2D and full 3D models on abdominal CT data.

10.2.2 Semi-Supervised Learning

FLARE22 includes an additional 2,000 unlabelled CT volumes. A semi-supervised approach — using the labelled 50 volumes for supervised training and the unlabelled 2,000 for consistency regularisation or pseudo-labelling — could dramatically expand the effective training set without the cost of manual annotation.

10.2.3 Uncertainty Estimation

The model currently produces a hard argmax prediction with no indication of confidence. Adding Monte Carlo dropout at inference (leaving the bottleneck Dropout2d active during evaluation) would produce per-voxel uncertainty estimates at negligible cost. Uncertainty maps are clinically useful — they highlight regions where the model is unsure and a radiologist should verify the segmentation.

10.2.4 Transformer-Based Backbones

Recent architectures such as Swin-UNETR replace convolutional encoders with Swin Transformers, capturing long-range spatial dependencies that convolutions miss. While these models require more data to train from scratch, fine-tuning a publicly available Swin-UNETR checkpoint on FLARE22 data would be feasible within the same hardware constraints as this project.

10.2.5 Multi-Dataset Pretraining

Training a shared backbone on several abdominal CT datasets (e.g., KiTS23, MSD-Liver, CHAOS) before fine-tuning on FLARE22 could improve robustness to scanner variability and organ co-occurrence patterns. This multi-dataset strategy is increasingly standard in large-scale medical imaging benchmarks and represents a natural next step for this work.

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